

Define a relation R on $\mathbb{Z}$ by xRy if and only if $5|(x + y)$

1. Two pairs in R

a. $\{7, 8\} \because 7 + 8 = 15 \text{ and } 5|15$

b. $\{9, 11\} \because 9 + 11 = 20 \text{ and } 5|20$

2. Two pairs not in R

a. $\{3, 4\} \because 3 + 4 = 7 \text{ and } 5 \nmid 7$

b. $\{4, 20\} \because 4 + 20 = 24 \text{ and } 5 \nmid 24$

3. Is R reflexive? Symmetric? Transitive?

Reflexive?

$$\{x, y\} = \{7, 8\}$$

$$\{x, x\} = \{7, 7\}$$

$$5 \mid (7 + 8) \text{ but } 5 \nmid (7 + 7)$$

Therefore R is not reflexive.

Symmetric?

Commutative Property shows that

$x + y = y + x$ which means if

$$\{x, y\} \in R \Rightarrow \{y, x\} \in R$$

Therefore R is Symmetric.

Transitive?

$$\{x, y\} = \{7, 8\} \in R \because 5 \mid (7 + 8)$$

$$\{y, z\} = \{8, 12\} \in R \because 5 \mid (8 + 12)$$

$$\{x, z\} = \{7, 12\} \notin R \because 5 \nmid (7 + 12)$$

Therefore R is not Transitive.